KCE PATHWAYS · GRADES 4–5 · UNIT 6: VOICES OF CHANGE · WEEK 26 · MATH

The Motor City Robot Rally — Teacher Lesson Plan & Answer Key

Learning objectives: Students evaluate numerical expressions with the order of operations (grouping symbols first, then multiplication and division from left to right, then addition and subtraction from left to right), write expressions from words with parentheses where they belong, work with brackets and braces from the inside out, place parentheses to make an equation true, and compare expressions by their structure without computing (for example, 4 × (387 + 29) is four times as large as 387 + 29).

Duration: 35–45 minutes
Materials: Computer or tablet with the lesson open · pencil · printed Pre & Post sheet

Lesson flow (as it appears on screen)

1. Slide 0 — Intro and name: meet Kavya (10), whose robotics team is competing at a robot rally in Detroit, Michigan. Coach Kasimir helps the team plan, her teammate Kendrick drives the robot, Ms. Kalinda, the head judge, keeps the official score, and her little brother Kamau tags along with a notebook full of questions. The student types first and last name so results are sent. Read the scoring rules together: 2 tries per question, Pathway Points only for a correct first try, and the answer plus the reason appear after the 2nd miss. Typed number answers ignore capital letters, commas in large numbers, dollar signs, extra spaces and a final period.
2. Slide 1 — Warm-Up, Part 1: Classifying Figures Review (4 pts, 2 each, Week 24 skill): a perimeter and a best name or angle classification. If a student misses both warm-up slides, redraw one figure and label its sides before moving on.
3. Slide 2 — Warm-Up, Part 2 (4 pts, 2 each): a side length from a perimeter, and lines of symmetry or perpendicular sides.
4. Slide 3 — Pencil Break (Pre): the student opens or prints the Pre/Post sheet (grade4-5/prepost/KCE_Math_PrePost_G4-5W26.pdf) and completes Page 1 (Pre-Lesson Warm-Up) off screen before any teaching.
5. Slide 4 — Why Robot Teams Need Expressions: nonfiction connection about Detroit, the “Motor City,” which hosted the FIRST Championship in 2018 and 2019 at Ford Field and its convention center on the Detroit River (then Cobo Center, now Huntington Place). FIRST stands for “For Inspiration and Recognition of Science and Technology.” A rally scoreboard shows that a judge’s score sheet is really a numerical expression. No questions.
6. Slide 5 — Picture It: The Order of Operations. Grouping symbols first; then multiply and divide from left to right; then add and subtract from left to right. 10 + 4 × 5 = 30 points (working straight left to right gives 70, which is wrong); 24 ÷ 4 × 2 = 12, not 3.
7. Slide 6 — Your Turn, Part 1: Evaluating Expressions (8 pts, 4 each): one typed evaluation and one multiple-choice value. Watch for students who work straight left to right.
8. Slide 7 — Your Turn, Part 2 (4 pts): a two-step score problem with Ms. Kalinda’s expression already written.
9. Slide 8 — Teach: Writing Expressions from Words. Coach Kasimir’s plans: “add 3 and 7, then multiply by 2” is 2 × (3 + 7); “subtract 5 from 40, then divide by 7” is (40 − 5) ÷ 7; “three times as large as 18 + 932” is 3 × (18 + 932), with no computing needed. Word clues: sum, difference, product, quotient.

10. Slide 9 — Your Turn, Part 1: Writing Expressions (8 pts, 4 each): type an expression (x or * for times and / for divide are accepted; spaces do not matter), and choose an expression or the words that match one.
11. Slide 10 — Your Turn, Part 2 (4 pts): write an expression from words, then evaluate it.
12. Slide 11 — Teach: Brackets, Braces, and Moving Parentheses. Work from the inside out: 2 × [5 + (9 − 3)] = 22. Same numbers, different values: 6 + 4 × 5 = 26, but (6 + 4) × 5 = 50; (24 − 4) × 3 = 60.
13. Slide 12 — Your Turn, Part 1: Grouping Symbols (8 pts, 4 each): evaluate from the inside out, and choose where the parentheses go.
14. Slide 13 — Your Turn, Part 2 (4 pts): add ONE pair of parentheses to make an equation true, then type the whole left side with the numbers in the same order.
15. Slide 14 — Vocabulary Round-Up: 8 flip cards (numerical expression, evaluate, order of operations, parentheses, brackets and braces, sum, product, quotient).
16. Slide 15 — Your Turn: Using the Words (6 pts, 3 each): one multiple-choice and one typed vocabulary item.
17. Slide 16 — Teach: Comparing Without Computing. Ms. Kalinda reads the structure: 4 × (387 + 29) is four times as large as 387 + 29; (906 − 14) ÷ 2 is half as large as 906 − 14; 6 × (450 + 25) is twice 3 × (450 + 25); but 10 × (78 − 6) is not the same as 10 × 78 − 6.
18. Slide 17 — Your Turn: Comparing Without Computing (6 pts, 3 each): “how many times as large?” and a choice that compares two expressions.
19. Slide 18 — Real World: The Qualifying Rounds (4 pts): a different situation in each set (top and low cones minus a penalty, three matches plus a teamwork bonus, or money left from $250 for parts).
20. Slide 19 — Real World: Pit Stop Planning (4 pts): 96 bonus points shared by 4 teams plus 3, 3 laps of a 14-foot and 9-foot practice track, or Kamau’s 7 + 5 × 8 that should have parentheses.
21. Slide 20 — Think It Through: Explain and Find the Mistake (10 pts, 5 each): a “which explanation is correct?” item and a “find the mistake” item. Ask the student to say why each wrong choice is wrong.
22. Slide 21 — Fix the Sentence, Part 1 (4 pts, 2 each): typed sentences checked exactly (quotation marks around dialogue, capital letters, commas with a city and state, and one grammar fix such as their/they’re or to/too). Straight or curly quotes and extra spaces are OK. The corrected sentence appears after the 2nd miss.
23. Slide 22 — Fix the Sentence, Part 2 (4 pts, 2 each): two more sentences (a series, a renaming phrase, or an opening word with a comma, and a verb that agrees with its real subject). Both serial-comma and no-serial-comma forms of a list are accepted.
24. Slide 23 — Match-Up Round (10 pts): 5 expressions matched to their values, word phrases matched to expressions, or result words (sum, product, quotient, difference, evaluate) matched to meanings, 2 pts each on the first try.
25. Slide 24 — Speed Trail (bonus XP only): 3 quick order-of-operations questions against the student’s own best time.
26. Slide 25 — Looking Back reflection (bonus XP only).
27. Slide 26 — Boss Battle: The Championship Match (8 pts, 4 each): a different situation in each set (two partner robots plus a 15-point bonus, Coach Kasimir’s parts order 3 × (12 + 4) + 10, or Kendrick’s drive routine repeated 4 times toward a 360-inch finish line): choose or read the expression, then evaluate it.
28. Slide 27 — Pencil Break (Post): the student completes Page 2 (Post-Lesson Challenge) of the same sheet (grade4-5/prepost/KCE_Math_PrePost_G4-5W26.pdf), plus the ★ Challenge if time allows, before seeing the score.
29. Slide 28 — Results (Rally Day Complete!): the score out of 100 is sent to KingCretot Experience. 85 or higher shows “Detroit Robot Rally Master!” with the congratulations message “Your scores passed every check at the Detroit robot rally!” and a big confetti burst; below 85 shows “Good Work at the Robot Rally!” with encouragement (“You are getting closer every time.”), and the next attempt brings a fresh question set (sets rotate 1, 2, 3). Both messages remind the student to finish the Post-Lesson Challenge sheet.

Lesson answer key with reasoning

Every scored item the student sees, in order, across all three rotating question sets. Each item allows 2 tries and earns Pathway Points only on a correct first try. The "why" is the same reasoning shown on screen after a second missed try.

WARM-UP: CLASSIFYING FIGURES REVIEW (2 POINTS EACH)

ID	Question	Answer	Why (explain to student)
r1_s0	Set 1: Figures (Week 24): the robot's base plate is a rhombus. One side is 7 inches long. What is its perimeter? Inches:	**28** (also accepted: 28 inches, 28 inch, 28 in)	All 4 sides of a rhombus are equal: 4 × 7 = 28 inches.
r2_s0	Set 1: Figures (Week 24): a tile on the competition field has 4 right angles and 4 equal sides. What is its BEST name? [Square / Rectangle that is not a square / Rhombus with no right angles / Trapezoid]	**Square**	Four right angles make it a rectangle, and four equal sides make it a rhombus. A shape that is both is a square, the most exact name.
r3_s0	Set 1: Figures (Week 24): a warning sign is an equilateral triangle with a perimeter of 39 centimeters. How long is each side? Centimeters:	**13** (also accepted: 13 centimeters, 13 centimeter, 13 cm)	The 3 sides are equal: 39 ÷ 3 = 13 centimeters.
r4_s0	Set 1: Symmetry (Week 24): how many lines of symmetry does a rectangle that is not a square have? Lines:	**2** (also accepted: 2 lines, 2 line, two)	Only the two folds across the middle make matching halves. The diagonals are not lines of symmetry.
r1_s1	Set 2: Figures (Week 24): a bumper is shaped like an isosceles triangle with two sides of 11 inches and a third side of 8 inches. What is its perimeter? Inches:	**30** (also accepted: 30 inches, 30 inch, 30 in)	Add all three sides: 11 + 11 + 8 = 30 inches.
r2_s1	Set 2: Figures (Week 24): which figure has EXACTLY one pair of parallel sides? [A trapezoid that is not a parallelogram / A rectangle / A rhombus / A square]	**A trapezoid that is not a parallelogram**	A rectangle, a rhombus, and a square are all parallelograms with 2 pairs of parallel sides. A trapezoid that is not a parallelogram has exactly 1 pair.
r3_s1	Set 2: Figures (Week 24): a gear guard is a regular hexagon with a perimeter of 54 millimeters. How long is each side? Millimeters:	**9** (also accepted: 9 millimeters, 9 millimeter, 9 mm)	A regular hexagon has 6 equal sides: 54 ÷ 6 = 9 millimeters.
r4_s1	Set 2: Figures (Week 24): how many pairs of perpendicular sides does a rectangle have? Pairs:	**4** (also accepted: 4 pairs, 4 pair, four)	A rectangle has 4 right angles, and each right angle is one pair of perpendicular sides: 4 pairs.
r1_s2	Set 3: Figures (Week 24): the robot's frame is a rectangle 18 inches long and 12 inches wide. What is its perimeter? Inches:	**60** (also accepted: 60 inches, 60 inch, 60 in)	Opposite sides are equal: 2 × 18 + 2 × 12 = 36 + 24 = 60 inches.
r2_s2	Set 3: Figures (Week 24): a triangle has angles of 95°, 45°, and 40°. How is it classified by its angles? [Obtuse / Acute / Right / Equilateral]	**Obtuse**	95° is more than 90°, so it is an obtuse triangle. One obtuse angle decides the name.
r3_s2	Set 3: Figures (Week 24): a square sticker for the robot has a perimeter of 64 centimeters. How long is each side? Centimeters:	**16** (also accepted: 16 centimeters, 16 centimeter, 16 cm)	A square has 4 equal sides: 64 ÷ 4 = 16 centimeters.

ID	Question	Answer	Why (explain to student)
r4_s2	Set 3: Symmetry (Week 24): how many lines of symmetry does a regular pentagon have? Lines:	**5** (also accepted: 5 lines, 5 line, five)	A regular polygon has as many lines of symmetry as it has sides. A pentagon has 5 sides, so it has 5 lines.

YOUR TURN: EVALUATING EXPRESSIONS (4 POINTS EACH)

ID	Question	Answer	Why (explain to student)
l1_s0	Set 1: Evaluate the expression: 18 + 6 × 4 Value:	**42**	Multiply first: 6 × 4 = 24. Then add: 18 + 24 = 42.
l2_s0	Set 1: What is the value of 36 ÷ 6 × 2? [12 / 3 / 24 / 72]	**12**	Divide and multiply from left to right: 36 ÷ 6 = 6, then 6 × 2 = 12. (3 comes from multiplying first, which skips the left-to-right rule.)
l3_s0	Set 1: Two steps: Kavya's robot drops 5 cubes worth 4 points each, loses 3 points for a penalty, and then earns a 15-point parking bonus. Ms. Kalinda writes 5 × 4 − 3 + 15. What is the score? Points:	**32** (also accepted: 32 points, 32 point, 32 pts)	Multiply first: 5 × 4 = 20. Then left to right: 20 − 3 = 17, and 17 + 15 = 32 points.
l1_s1	Set 2: Evaluate the expression: 50 − 8 × 5 Value:	**10**	Multiply first: 8 × 5 = 40. Then subtract: 50 − 40 = 10.
l2_s1	Set 2: What is the value of 48 ÷ 4 ÷ 2? [6 / 24 / 96 / 12]	**6**	Divide from left to right: 48 ÷ 4 = 12, then 12 ÷ 2 = 6. (24 comes from doing 4 ÷ 2 first.)
l3_s1	Set 2: Two steps: Kendrick's robot places 7 cones worth 3 points each and 2 cubes worth 5 points each. Ms. Kalinda writes 7 × 3 + 2 × 5. What is the score? Points:	**31** (also accepted: 31 points, 31 point, 31 pts)	Do both multiplications first: 7 × 3 = 21 and 2 × 5 = 10. Then add: 21 + 10 = 31 points.
l1_s2	Set 3: Evaluate the expression: (14 + 6) × 3 − 9 Value:	**51**	Parentheses first: 14 + 6 = 20. Multiply: 20 × 3 = 60. Subtract: 60 − 9 = 51.
l2_s2	Set 3: What is the value of 30 − 12 + 8? [26 / 10 / 50 / 34]	**26**	Add and subtract from left to right: 30 − 12 = 18, then 18 + 8 = 26. (10 comes from adding 12 + 8 first.)
l3_s2	Set 3: Two steps: Kavya's team scores 24 points in each of 3 rounds. Then a 60-point bonus is shared equally by 4 teams. Ms. Kalinda writes 3 × 24 + 60 ÷ 4 for the team's total. What is it? Points:	**87** (also accepted: 87 points, 87 point, 87 pts)	Multiply and divide first: 3 × 24 = 72 and 60 ÷ 4 = 15. Then add: 72 + 15 = 87 points.

YOUR TURN: WRITING EXPRESSIONS (4 POINTS EACH)

ID	Question	Answer	Why (explain to student)
k1_s0	Set 1: Write an expression for: add 12 and 9, then multiply by 4. (Do not evaluate it.)	**4×(12+9)** (also accepted: 4x(12+9), 4*(12+9), (12+9)×4, (12+9)x4, (12+9)*4, 4×(9+12), 4x(9+12), 4*(9+12), (9+12)×4, (9+12)x4, (9+12)*4)	The words say to add first, so 12 + 9 goes inside parentheses, then the sum is multiplied by 4: 4 × (12 + 9).

ID	Question	Answer	Why (explain to student)
k2_s0	Set 1: Which expression means “three times as large as 18 + 932”? [3 × (18 + 932) / 3 × 18 + 932 / 18 + 932 + 3 / (18 + 932) ÷ 3]	**3 × (18 + 932)**	The whole sum, 18 + 932, is made three times as large, so parentheses must hold the sum: 3 × (18 + 932). In 3 × 18 + 932, only the 18 is tripled.
k3_s0	Set 1: Two steps: Kamau reads from his notebook, “Add 16 and 8, then divide by 3.” Write the expression and evaluate it. What is the value? Value:	**8**	The expression is (16 + 8) ÷ 3. Parentheses first: 16 + 8 = 24. Then 24 ÷ 3 = 8.
k1_s1	Set 2: Write an expression for: subtract 6 from 50, then divide by 4. (Do not evaluate it.)	**(50−6)÷4** (also accepted: (50-6)÷4, (50−6)/4, (50-6)/4)	Subtract 6 from 50 means start with 50: 50 − 6. That step comes first, so it goes in parentheses: (50 − 6) ÷ 4.
k2_s1	Set 2: Which words describe (603 + 74) ÷ 2? [Half of the sum of 603 and 74 / 603 plus half of 74 / Twice the sum of 603 and 74 / The sum of 603, 74, and 2]	**Half of the sum of 603 and 74**	The parentheses make 603 + 74 one sum, and dividing by 2 takes half of it.
k3_s1	Set 2: Two steps: Coach Kasimir buys 4 repair kits. Each kit has 12 bolts and 8 nuts. Write an expression for the total number of parts and evaluate it. Parts:	**80** (also accepted: 80 parts, 80 part)	One kit has 12 + 8 parts, so 4 kits have 4 × (12 + 8) = 4 × 20 = 80 parts.
k1_s2	Set 3: Write an expression for: multiply 7 by the sum of 15 and 25. (Do not evaluate it.)	**7×(15+25)** (also accepted: 7x(15+25), 7*(15+25), (15+25)×7, (15+25)x7, (15+25)*7, 7×(25+15), 7x(25+15), 7*(25+15), (25+15)×7, (25+15)x7, (25+15)*7)	The sum of 15 and 25 is one group, so it goes in parentheses, and 7 multiplies the whole group: 7 × (15 + 25).
k2_s2	Set 3: Which expression means “subtract 9 from 45, then multiply by 6”? [(45 − 9) × 6 / 45 − 9 × 6 / (9 − 45) × 6 / 45 − (9 × 6)]	**(45 − 9) × 6**	Subtract 9 from 45 means 45 − 9, and "then" means it happens first, so it goes in parentheses before multiplying by 6.
k3_s2	Set 3: Two steps: Ms. Kalinda says, “Subtract 25 from 100, then divide by 5.” Write the expression and evaluate it. What is the value? Value:	**15**	The expression is (100 − 25) ÷ 5. Parentheses first: 100 − 25 = 75. Then 75 ÷ 5 = 15.

YOUR TURN: GROUPING SYMBOLS (4 POINTS EACH)

ID	Question	Answer	Why (explain to student)
e1_s0	Set 1: Evaluate from the inside out: 3 × [20 − (4 + 6)] Value:	**30**	Parentheses: 4 + 6 = 10. Brackets: 20 − 10 = 10. Then 3 × 10 = 30.
e2_s0	Set 1: Where do the parentheses go so that 8 + 2 × 5 = 50? [(8 + 2) × 5 / 8 + (2 × 5) / (8) + 2 × 5 / 8 + 2 × (5)]	**(8 + 2) × 5**	(8 + 2) × 5 = 10 × 5 = 50. Every other choice still multiplies 2 × 5 first and gives 18.
e3_s0	Set 1: Two steps: add one pair of parentheses so that 4 + 6 × 3 − 1 = 29. Test your choice, then type the left side.	**(4+6)×3−1** (also accepted: (4+6)×3-1, (4+6)x3−1, (4+6)x3-1, (4+6)*3−1, (4+6)*3-1)	Try (4 + 6) × 3 − 1: 10 × 3 = 30, and 30 − 1 = 29. Other placements give 16 or 21.
e1_s1	Set 2: Evaluate from the inside out: [(7 + 5) × 3] ÷ 4 Value:	**9**	Parentheses: 7 + 5 = 12. Brackets: 12 × 3 = 36. Then 36 ÷ 4 = 9.

ID	Question	Answer	Why (explain to student)
e2_s1	Set 2: Which placement of parentheses makes 36 ÷ 6 − 2 equal 9? [36 ÷ (6 − 2) / (36 ÷ 6) − 2 / (36) ÷ 6 − 2 / 36 ÷ 6 − (2)]	**36 ÷ (6 − 2)**	36 ÷ (6 − 2) = 36 ÷ 4 = 9. The other choices all divide 36 ÷ 6 first and give 4.
e3_s1	Set 2: Two steps: add one pair of parentheses so that 30 − 10 ÷ 5 + 3 = 7. Test your choice, then type the left side.	**(30−10)÷5+3** (also accepted: (30-10)÷5+3, (30−10)/5+3, (30-10)/5+3)	Try (30 − 10) ÷ 5 + 3: 20 ÷ 5 = 4, and 4 + 3 = 7. Without parentheses the value is 30 − 2 + 3 = 31.
e1_s2	Set 3: Evaluate from the inside out: {40 − [2 × (3 + 5)]} ÷ 6 Value:	**4**	Parentheses: 3 + 5 = 8. Brackets: 2 × 8 = 16. Braces: 40 − 16 = 24. Then 24 ÷ 6 = 4.
e2_s2	Set 3: Which expression has the GREATEST value? [(5 + 3) × 4 / 5 + 3 × 4 / 5 × 3 + 4 / 5 + (3 × 4)]	**(5 + 3) × 4**	(5 + 3) × 4 = 32. The others: 5 + 3 × 4 = 17, 5 × 3 + 4 = 19, and 5 + (3 × 4) = 17.
e3_s2	Set 3: Two steps: add one pair of parentheses so that 2 × 9 + 3 × 4 = 96. Test your choice, then type the left side.	**2×(9+3)×4** (also accepted: 2x(9+3)x4, 2*(9+3)*4)	Try 2 × (9 + 3) × 4: 9 + 3 = 12, then 2 × 12 = 24, and 24 × 4 = 96. Other placements give 30, 42, or 84.

YOUR TURN: USING THE WORDS (3 POINTS EACH)

ID	Question	Answer	Why (explain to student)
v1_s0	Set 1: From the lesson: “To evaluate an expression means to find its value.” Which sentence shows someone evaluating? [Kendrick works out that 6 + 2 × 3 equals 12 / Kavya writes 6 + 2 × 3 on the whiteboard / Kamau copies the expression into his notebook / Coach Kasimir reads the expression out loud]	**Kendrick works out that 6 + 2 × 3 equals 12**	Evaluating means finding the value. 6 + 2 × 3 = 6 + 6 = 12. Writing, copying, or reading an expression does not find its value.
v2_s0	Set 1: Type your answer: in 7 × 8 = 56, the number 56 is the ___. (one word)	**product** (also accepted: the product; with “a” or “an” or in the plural)	The answer to a multiplication is the product.
v1_s1	Set 2: From the lesson: “Brackets [ ] and braces { } are grouping symbols that work like bigger parentheses.” In 4 × [10 − (2 + 3)], which step comes FIRST? [2 + 3 / 4 × 10 / 10 − 2 / 4 × 3]	**2 + 3**	Work from the inside out: the parentheses (2 + 3) sit inside the brackets, so they come first.
v2_s1	Set 2: Type your answer: in 45 ÷ 9 = 5, the number 5 is the ___. (one word)	**quotient** (also accepted: the quotient; with “a” or “an” or in the plural)	The answer to a division is the quotient.
v1_s2	Set 3: From the lesson: “A numerical expression is a group of numbers and operation symbols that names one value.” Which of these is a numerical expression? [9 × (4 + 1) / the robot / + × − / Detroit, Michigan]	**9 × (4 + 1)**	9 × (4 + 1) has numbers joined by operation symbols, and it names one value, 45.
v2_s2	Set 3: Type your answer: in 26 + 14 = 40, the number 40 is the ___. (one word)	**sum** (also accepted: the sum; with “a” or “an” or in the plural)	The answer to an addition is the sum.

YOUR TURN: COMPARING WITHOUT COMPUTING (3 POINTS EACH)

ID	Question	Answer	Why (explain to student)
t1_s0	Set 1: Without computing: 7 × (2,418 + 365) is how many times as large as 2,418 + 365? Times as large:	**7** (also accepted: 7 times, seven)	The whole sum 2,418 + 365 is multiplied by 7, so the expression is 7 times as large.

ID	Question	Answer	Why (explain to student)
t2_s0	Set 1: Which expression is TWICE as large as 615 − 48? [2 × (615 − 48) / 2 × 615 − 48 / (615 − 48) ÷ 2 / 615 − 48 + 2]	**2 × (615 − 48)**	Only 2 × (615 − 48) doubles the whole difference. 2 × 615 − 48 doubles just the 615.
t1_s1	Set 2: Without computing: 9 × (814 − 76) is how many times as large as 3 × (814 − 76)? Times as large:	**3** (also accepted: 3 times, three)	Both multiply the same group, 814 − 76. 9 is 3 times as large as 3, so the first expression is 3 times as large.
t2_s1	Set 2: Which statement correctly compares (12 + 38) × 6 and 12 + 38 × 6? [The first is greater: 300 compared with 240 / They are equal because they use the same numbers / The second is greater: 300 compared with 240 / The first is greater: 300 compared with 50]	**The first is greater: 300 compared with 240**	(12 + 38) × 6 = 50 × 6 = 300. 12 + 38 × 6 = 12 + 228 = 240. So the first is greater.
t1_s2	Set 3: Without computing: 8 × (93 + 47) is how many times as large as 2 × (93 + 47)? Times as large:	**4** (also accepted: 4 times, four)	Both multiply the same group, 93 + 47. 8 ÷ 2 = 4, so the first expression is 4 times as large.
t2_s2	Set 3: Which expression is NOT three times as large as 147 + 59? [3 × 147 + 59 / 3 × (147 + 59) / (147 + 59) × 3 / (147 + 59) + (147 + 59) + (147 + 59)]	**3 × 147 + 59**	In 3 × 147 + 59, only 147 is tripled; the 59 is added once. The other three all triple the whole sum.

REAL WORLD: THE QUALIFYING ROUNDS (4 POINTS EACH)

ID	Question	Answer	Why (explain to student)
w1_s0	Set 1: *The Situation In the first qualifying round, a cone on the top shelf is worth 5 points and a cone on the low shelf is worth 2 points. Kavya's robot places 6 top cones and 9 low cones. Then it bumps a wall and loses 10 points.* Two steps: what is the robot's score? (6 × 5 + 9 × 2 − 10) Points:	**38** (also accepted: 38 points, 38 point, 38 pts)	Multiply first: 6 × 5 = 30 and 9 × 2 = 18. Then left to right: 30 + 18 = 48, and 48 − 10 = 38 points.
w1_s1	Set 2: *The Situation Kendrick's team scores 45 points in each of 3 qualifying matches. After the last match, Ms. Kalinda adds one 12-point teamwork bonus to the team's total.* Two steps: which expression and value give the team's final total? [3 × 45 + 12 = 147 / 3 × (45 + 12) = 171 / 45 + 3 × 12 = 81 / 3 × 45 + 12 = 171]	**3 × 45 + 12 = 147**	The bonus is added only once, after the 3 matches: 3 × 45 + 12 = 135 + 12 = 147. 3 × (45 + 12) would give the bonus 3 times.
w1_s2	Set 3: *The Situation Coach Kasimir has $250 to spend on parts. He buys 4 motors that cost $38 each and 2 sensors that cost $24 each.* Two steps: how much money does he have left? (250 − (4 × 38 + 2 × 24)) Dollars:	**50** (also accepted: 50 dollars, 50.00, 50 dollar)	Inside the parentheses: 4 × 38 = 152 and 2 × 24 = 48, so 152 + 48 = 200. Then 250 − 200 = $50.

REAL WORLD: PIT STOP PLANNING (4 POINTS EACH)

ID	Question	Answer	Why (explain to student)
w2_s0	Set 1: *The Situation Ms. Kalinda shares 96 bonus points equally among 4 teams. Then Kavya's team earns 3 more points for a clean match with no penalties.* Two steps: which expression gives the bonus points Kavya's team ends with, and what is its value? [96 ÷ 4 + 3 = 27 / 96 ÷ (4 + 3) = 27 / (96 + 3) ÷ 4 = 27 / 96 ÷ 4 × 3 = 72]	**96 ÷ 4 + 3 = 27**	Share first, then add: 96 ÷ 4 = 24, and 24 + 3 = 27. 96 ÷ (4 + 3) and (96 + 3) ÷ 4 do not equal 27, and 96 ÷ 4 × 3 triples the share instead of adding 3.
w2_s1	Set 2: *The Situation On the practice track, one lap has 2 straight sides of 14 feet and 2 curved ends of 9 feet. Kendrick drives the robot around the track 3 times.* Two steps: how many feet does the robot travel? (3 × (2 × 14 + 2 × 9)) Feet:	**138** (also accepted: 138 feet, 138 foot, 138 ft)	One lap: 2 × 14 + 2 × 9 = 28 + 18 = 46 feet. Three laps: 3 × 46 = 138 feet.
w2_s2	Set 3: *The Situation Kamau wants to write "add 7 and 5, then multiply by 8." In his notebook he writes 7 + 5 × 8.* Two steps: what value does Kamau's expression give, and what value should it give? [His gives 47; (7 + 5) × 8 gives 96 / His gives 96; that is correct / His gives 47; (7 + 5) × 8 gives 60 / His gives 60; (7 + 5) × 8 gives 96]	**His gives 47; (7 + 5) × 8 gives 96**	Without parentheses, multiply first: 7 + 40 = 47. The words say add first, so (7 + 5) × 8 = 12 × 8 = 96.

THINK IT THROUGH: EXPLAIN AND FIND THE MISTAKE (5 POINTS EACH)

ID	Question	Answer	Why (explain to student)
x1_s0	Set 1: Which explanation is correct for 20 − 2 × 5? [Multiply first: 2 × 5 = 10, then 20 − 10 = 10 / Work left to right: 20 − 2 = 18, then 18 × 5 = 90 / Subtract first because subtraction comes first in the expression / Add all the numbers: 20 + 2 + 5 = 27]	**Multiply first: 2 × 5 = 10, then 20 − 10 = 10**	Multiplication comes before subtraction, so 2 × 5 = 10 first, and 20 − 10 = 10.
x2_s0	Set 1: Find the mistake: Kendrick evaluates 48 ÷ 6 × 2 like this: 6 × 2 = 12, so 48 ÷ 12 = 4. What should he fix? [Multiply and divide from left to right: 48 ÷ 6 = 8, then 8 × 2 = 16 / Nothing. The answer 4 is correct / He should add first: 48 + 6 + 2 = 56 / He should always multiply before dividing]	**Multiply and divide from left to right: 48 ÷ 6 = 8, then 8 × 2 = 16**	Multiplication does not come before division. They are partners, done in order from left to right: 48 ÷ 6 = 8, and 8 × 2 = 16.
x1_s1	Set 2: Which explanation is correct? Does 5 × (8 + 2) have the same value as 5 × 8 + 2? [No. The parentheses mean add first: 5 × 10 = 50, but 5 × 8 + 2 = 42 / Yes, because they use the same numbers / Yes, both equal 50 / No. Both equal 42, but they look different]	**No. The parentheses mean add first: 5 × 10 = 50, but 5 × 8 + 2 = 42**	Parentheses change the order. 5 × (8 + 2) = 50, and 5 × 8 + 2 = 40 + 2 = 42.

ID	Question	Answer	Why (explain to student)
x2_s1	Set 2: Find the mistake: Kavya writes “four times as large as 26 + 9” as 4 × 26 + 9. What should she fix? [Add parentheses: 4 × (26 + 9), so the whole sum is multiplied / Nothing. 4 × 26 + 9 is correct / Change it to 4 + 26 + 9 / Change it to (4 × 26) + 9]	**Add parentheses: 4 × (26 + 9), so the whole sum is multiplied**	Without parentheses, only 26 is multiplied by 4. The words mean the whole sum is four times as large, so write 4 × (26 + 9).
x1_s2	Set 3: Which explanation is correct for 2 × [3 + (10 − 4)]? [Parentheses first: 10 − 4 = 6; then brackets: 3 + 6 = 9; then 2 × 9 = 18 / Left to right: 2 × 3 = 6, then 6 + 10 = 16, then 16 − 4 = 12 / Brackets first: 2 × 3 = 6, then 6 + 6 = 12 / Ignore the grouping symbols: 2 + 3 + 10 + 4 = 19]	**Parentheses first: 10 − 4 = 6; then brackets: 3 + 6 = 9; then 2 × 9 = 18**	Work from the inside out. The parentheses sit inside the brackets: 10 − 4 = 6, then 3 + 6 = 9, then 2 × 9 = 18.
x2_s2	Set 3: Find the mistake: Kamau says 30 − 10 + 5 = 15, because you always add before you subtract. What should he fix? [Add and subtract from left to right: 30 − 10 = 20, then 20 + 5 = 25 / Nothing. 15 is correct / Always subtract first, so the answer is 15 / Multiply first: 30 × 10 + 5 = 305]	**Add and subtract from left to right: 30 − 10 = 20, then 20 + 5 = 25**	Addition does not come before subtraction. They are partners, done from left to right: 30 − 10 = 20, and 20 + 5 = 25.

FIX THE SENTENCE (2 POINTS EACH)

ID	Question	Answer	Why (explain to student)
s1_s0	Set 1: Fix this sentence: kavya asked, do we multiply first	**Kavya asked, "Do we multiply first?"**	Capitalize the name Kavya, put quotation marks around her exact words, capitalize the first word of the quote, and put the question mark inside the closing quotation mark.
s2_s0	Set 1: Fix this sentence: The rally is at huntington place in detroit, michigan.	**The rally is at Huntington Place in Detroit, Michigan.**	Names of buildings, cities, and states are proper nouns, so capitalize Huntington Place, Detroit, and Michigan.
s3_s0	Set 1: Fix this sentence: The robot needs motors sensors and wheels.	**The robot needs motors, sensors, and wheels.** (also accepted: The robot needs motors, sensors and wheels.)	Use commas to separate the items in a series: motors, sensors, and wheels.
s4_s0	Set 1: Fix this sentence: each of the robots have four wheels	**Each of the robots has four wheels.**	Capitalize the first word. The subject is each (one robot at a time), so use has, not have. End with a period.
s1_s1	Set 2: Fix this sentence: coach kasimir said, check the parentheses first	**Coach Kasimir said, "Check the parentheses first."** (also accepted: Coach Kasimir said, "Check the parentheses first!")	Capitalize Coach Kasimir, a title used with a name. Put quotation marks around his exact words, capitalize the first word of the quote, and put the end mark inside the closing quotation mark.
s2_s1	Set 2: Fix this sentence: The teams checked they're scores twice.	**The teams checked their scores twice.**	Their shows ownership: the scores belong to the teams. They're means they are.

ID	Question	Answer	Why (explain to student)
s3_s1	Set 2: Fix this sentence: Ms. Kalinda the head judge added the points.	**Ms. Kalinda, the head judge, added the points.**	The head judge describes Ms. Kalinda. Put a comma before and after a phrase like this that renames the person.
s4_s1	Set 2: Fix this sentence: the wheels on the robot is loose	**The wheels on the robot are loose.**	Capitalize the first word. The subject is wheels (plural), not robot, so use are, not is. End with a period.
s1_s2	Set 3: Fix this sentence: kendrick shouted, our robot scored again	**Kendrick shouted, "Our robot scored again!"** (also accepted: Kendrick shouted, "Our robot scored again.")	Capitalize the name Kendrick, put quotation marks around his exact words, capitalize the first word of the quote, and put the end mark inside the closing quotation mark.
s2_s2	Set 3: Fix this sentence: Kamau wants to build a robot to.	**Kamau wants to build a robot too.**	Too means also: Kamau wants to build a robot, also. The first to (to build) is correct.
s3_s2	Set 3: Fix this sentence: No we do not add before we multiply.	**No, we do not add before we multiply.**	Put a comma after an opening word like Yes, No, or Well.
s4_s2	Set 3: Fix this sentence: the points from each round is added together	**The points from each round are added together.**	Capitalize the first word. The subject is points (plural), not round, so use are, not is. End with a period.

MATCH-UP ROUND (2 POINTS PER FIRST-TRY MATCH)

ID	Question	Answer	Why (explain to student)
match_s0_1	Set 1: Match 6 + 4 × 2	**14**	2 points if matched on the first try.
match_s0_2	Set 1: Match (6 + 4) × 2	**20**	2 points if matched on the first try.
match_s0_3	Set 1: Match 18 − 6 ÷ 3	**16**	2 points if matched on the first try.
match_s0_4	Set 1: Match (18 − 6) ÷ 3	**4**	2 points if matched on the first try.
match_s0_5	Set 1: Match 2 × [1 + (5 − 1)]	**10**	2 points if matched on the first try.
match_s1_1	Set 2: Match add 5 and 3, then multiply by 2	**(5 + 3) × 2**	2 points if matched on the first try.
match_s1_2	Set 2: Match multiply 5 by 3, then add 2	**5 × 3 + 2**	2 points if matched on the first try.
match_s1_3	Set 2: Match subtract 3 from 5, then multiply by 2	**(5 − 3) × 2**	2 points if matched on the first try.
match_s1_4	Set 2: Match half of the sum of 5 and 3	**(5 + 3) ÷ 2**	2 points if matched on the first try.
match_s1_5	Set 2: Match add 5 to the product of 3 and 2	**5 + 3 × 2**	2 points if matched on the first try.
match_s2_1	Set 3: Match the answer to an addition	**sum**	2 points if matched on the first try.
match_s2_2	Set 3: Match the answer to a multiplication	**product**	2 points if matched on the first try.
match_s2_3	Set 3: Match the answer to a division	**quotient**	2 points if matched on the first try.
match_s2_4	Set 3: Match the answer to a subtraction	**difference**	2 points if matched on the first try.
match_s2_5	Set 3: Match to find the value of an expression	**evaluate**	2 points if matched on the first try.

BOSS BATTLE: THE CHAMPIONSHIP MATCH (4 POINTS EACH)

ID	Question	Answer	Why (explain to student)
b1_s0	Set 1: *Kavya and Kendrick in the Final In the final match, two robots play together as partners. Kavya's robot places 8 cones worth 3 points each and 4 cubes worth 5 points each. The partner robot scores exactly the same. Then Ms. Kalinda adds one 15-point bonus to the partners' total.* Which expression gives the partners' total score? [2 × (8 × 3 + 4 × 5) + 15 / 2 × 8 × 3 + 4 × 5 + 15 / 8 × 3 + 4 × 5 + 15 / 2 × (8 × 3 + 4 × 5 + 15)]	**2 × (8 × 3 + 4 × 5) + 15**	One robot scores 8 × 3 + 4 × 5. Two robots score twice that, so it goes in parentheses times 2. The bonus is added once, outside the parentheses.
b2_s0	Set 1: Two steps: what is the partners' total score? Points:	**103** (also accepted: 103 points, 103 point, 103 pts)	One robot: 8 × 3 + 4 × 5 = 24 + 20 = 44. Two robots: 2 × 44 = 88. Add the bonus: 88 + 15 = 103 points.
b1_s1	Set 2: *Coach Kasimir Restocks the Pit Before the final, Coach Kasimir orders 3 parts kits. Each kit has 12 bolts and 4 motor mounts. He also orders 10 extra bolts. His order form says 3 × (12 + 4) + 10.* Which words match 3 × (12 + 4) + 10? [Three times the number of parts in one kit, plus 10 extra bolts / Three times the number of bolts, plus 4 mounts and 10 bolts / The parts in one kit, plus 3 times 10 bolts / Three times the parts in one kit plus 10 bolts, all multiplied by 3 again]	**Three times the number of parts in one kit, plus 10 extra bolts**	The parentheses hold one kit, 12 + 4. The 3 multiplies that kit. The 10 extra bolts are added once at the end.
b2_s1	Set 2: Two steps: how many parts does Coach Kasimir order in all? Parts:	**58** (also accepted: 58 parts, 58 part)	One kit: 12 + 4 = 16 parts. Three kits: 3 × 16 = 48. Add the extra bolts: 48 + 10 = 58 parts.
b1_s2	Set 3: *Kendrick Tests the Drive Routine In its routine, the robot drives forward 120 inches and then backs up 35 inches. It repeats this routine 4 times. The finish line is 360 inches from the start.* Which expression shows how far from the start the robot ends? [4 × (120 − 35) / 4 × 120 − 35 / 120 − 35 × 4 / 4 × 120 − 4 + 35]	**4 × (120 − 35)**	Each routine moves the robot 120 − 35 inches forward. Four routines: 4 × (120 − 35). In 4 × 120 − 35, the robot would back up only once.
b2_s2	Set 3: Two steps: how many more inches must the robot go to reach the finish line? Inches:	**20** (also accepted: 20 inches, 20 inch, 20 in)	One routine: 120 − 35 = 85 inches. Four routines: 4 × 85 = 340 inches. Still to go: 360 − 340 = 20 inches.

Pre & Post sheet answer guide

Many items are open-ended. Grade for genuine understanding and complete sentences, not exact wording. Sheet: grade4-5/prepost/KCE_Math_PrePost_G4-5W26.pdf.

PRE-LESSON WARM-UP

#	Prompt	Model answer / grading note
1	Review: regular hexagon wheel guard with 6-centimeter sides	a) 6 × 6 = 36 centimeters b) 6 lines of symmetry (a regular polygon has as many lines as sides).

#	Prompt	Model answer / grading note
2	3 cones worth 4 points each: multiplication sentence and total	3 × 4 = 12 points
3	Kamau says 2 + 3 × 4 = 20; Kendrick says 14	Kendrick is right: multiply first, 3 × 4 = 12, then 2 + 12 = 14. Kamau added first. Accept any reasonable thinking before the lesson.
4	What the parentheses tell you to do first in (5 + 5) × 3	Add inside the parentheses first: 5 + 5 = 10, then 10 × 3 = 30.
5	Expression for “add 6 and 2, then multiply by 5”	(6 + 2) × 5

POST-LESSON CHALLENGE

#	Prompt	Model answer / grading note
1	a) 40 − 6 × 5 b) 72 ÷ 8 × 3 c) 2 × [18 − (5 + 4)]	a) 40 − 30 = 10 b) 9 × 3 = 27 (left to right) c) 2 × [18 − 9] = 2 × 9 = 18
2	Write, do not evaluate: a) subtract 8 from 50, then divide by 6 b) four times as large as 327 + 85	a) (50 − 8) ÷ 6 b) 4 × (327 + 85)
3	9 cones at 3 points, 5 cubes at 4 points, minus a 6-point penalty	9 × 3 + 5 × 4 − 6 = 27 + 20 − 6 = 41 points.
4	Kendrick says 36 − 12 + 4 = 20	Model: Kendrick added before he subtracted. Addition and subtraction are done from left to right, so 36 − 12 = 24 comes first. Then 24 + 4 = 28, so the correct value is 28.

★ CHALLENGE

#	Prompt	Model answer / grading note
1	One pair of parentheses: a) 3 + 5 × 4 − 2 = 30 b) 48 ÷ 4 + 2 × 3 = 24	a) (3 + 5) × 4 − 2 = 32 − 2 = 30 b) 48 ÷ (4 + 2) × 3 = 8 × 3 = 24